

برای هریک از 4 بخش عنوان، چکیده، مقدمه، جمع بندی و نتیجه گیری مقاله ساده شده زیر، الف در مجموع 20 مورد و به صورت تیتروار در صفحه دوم همین فایل مشخص کنید که هر بخش از لحاظ ساختاری باید شامل چه قسمتهایی باشد. ب برای هر قسمت از مقاله کوتاه شده ضمیمه، با علامت گذاری یا رنگ آمیزی در همین مقاله، فقط مشخص کنید از مواردی که در قسمت الف نوشتید کدام بخشها در این مقاله وجود دارند. مثلاً اگر بخشی قسمت X را دارد، می توانید جملاتی که مربوط به X هستند را فقط روی متن مقاله علامت گذاری کنید.

An Interactive Approach to Teaching Git Version Control System

Abstract

A1: Although the Git version control system is widely used in software engineering, it has been observed that most Computer Science and Software Engineering students do not have the necessary knowledge and practices to use Git. **A2:** To address this issue, we have prepared a Git and GitHub training program consisting of four sessions as a part of the Object-Oriented Software Engineering course where junior students utilized these tools for their term projects. **A3:** The program was conducted in three academic terms for a total of 258 students. To evaluate the effectiveness of the training

sessions, we have conducted two surveys, before 224 respondents and after 200 respondents the program. **A4:** According to the survey results, the number of students considering themselves insufficient to use the tools for their projects decreased from 67% to 9% after the training program. **A5:** Additionally, the majority of the students found the lectures and laboratory assignments beneficial.

Introduction

I1: Version control systems allow software developers to track changes to source code over time and recall a specific version later if needed. **I2:** These systems are generally divided into two categories: centralized and distributed. In centralized version control systems e.g., Subversion¹, there is a single server that contains all the project files, and users can only work on the latest snapshot of the project. One downside of such systems is that there is a single point of failure, meaning that the project could be lost if something goes wrong with the server e.g., database corruption. In the last decade, many developers moved their projects to distributed version control systems such as Git² where each user stores the complete history of a project, making it immune to server related issues. Git is currently one of the most popular version control systems, and it was used by 87.2% of responding developers in the annual developer survey carried out by Stack Overflow in 2018 [1]. **I3:** However, despite the trend, it has been noticed that most students lack the necessary knowledge and practice to use Git. This is because version control systems are not adequately addressed in typical Computer Science or Software Engineering curricula [2, 3]. **I4:** To this end, we prepared an educational training program for junior Computer Science students as a part of the Object-Oriented Software Engineering course in three academic terms which are referred to as the first, second and third iterations of the program. In this course, students

are divided into 5-member teams to develop a software project using object-oriented principles collaboratively. Moreover, we made it mandatory for students to use Git and GitHub³ throughout their term projects. The training program aimed to introduce frequently used Git and GitHub concepts to students and prepare them for their software engineering careers. We followed a similar structure for the training programs in all iterations, starting with a pre-training survey to see the students' level of experience in Git and GitHub. Then, we delivered a lecture introducing the general concepts of version control systems and GitHub. Next, we conducted a live session to show the Git instructions in more detail. Before executing the last session, we allowed the students to learn more about and practice the tools remotely for a week through online materials as well as the lecture slides and tutorial videos we have prepared. Afterward, as the last session of the program, the students were given a graded 5% of the total course grade laboratory assignment, where they had to execute a specific scenario using Git/GitHub operations. At the end of the program, we conducted a post-training survey to assess the efficacy of the training sessions. **I5:** The rest of the paper is organized as follows. Section 2 reviews the literature regarding the use of Git and GitHub in education and the teaching of these tools. Section 3 describes the training process in detail, while Section 4 presents the results and observations. Section 5 and 6 discuss the potential threats to the validity of the study and lessons learned, respectively. Finally, Section 7 concludes the study.

Conclusion

C1: In this study, we have demonstrated how we conducted a Git and GitHub training program for third-year undergraduate students in three academic terms as a part of the Object-Oriented Software Engineering course. The motivation for conducting the training program came from the lack of representation regarding the usage of version control systems in Computer Science and Software Engineering curricula. **C2:** We designed the program to address our learning objectives directly.

During the program, we delivered lectures and shared our pre-prepared YouTube videos with the students so that they could gain practical experience. **C3:** To evaluate the students' knowledge, an interactive laboratory assignment was conducted in a virtual environment. **C2:** We shared the lecture slides, YouTube tutorials, and assignment materials publicly, making them accessible for the Computer Science education community. **In**

order to assess the effectiveness of the tutorial and laboratory assignment, pre and post-training surveys were carried out. C4: Comparing the combined survey results in both iterations, in the beginning, 63% of the students used Git previously; however, after the tutorials and the laboratory assignment, 92% of the students stated that they learned Git prominently. The percentage of students who were confident in using Git and GitHub for their term projects rose from 33% to 91%, which indicates that the

training program we designed for teaching Git and GitHub was successful. C5: In the future iterations of the program, we plan to make the laboratory session a collaborative activity to allow students to have an experience closer to real-life scenarios. Moreover, we are planning to expand the scope of the laboratory assignment to include more GitHub operations such as creating pull requests, opening issues, etc.

عنوان Title

1. چی؟ خروجی/محصول/موضوع اصلی مقاله را روشن کند روش/مدل/چارچوب/سیستم....
 2. چرا؟ هدف/مسئله/شکاف را نشان دهد این کار برای حل چه چیزی است؟
 3. چگونه؟ رویکرد/ویژگی متمایز یا روش کلیدی را بیاورد.
 4. کجا؟ زمینه/دامنه کاربرد را در صورت نیاز مشخص کند.
- عنوان اجزای 1 و 2 و 3 را دارد. جزء 4 هم به صورت کلی ذکر شده که teaching است میشد صریح تر هم ذکر کنه. توی 1 اومده گفته که Approach است توی 2 گفته که برای version control است. توی 3 هم گفته Interactive است توی 4 هم گفته teaching است.
- عنوان باید مختصر، نه خیلی تخصصی، بدون قیود توصیفی-کیفی، نه خیلی کوتاه/بلند، و همخوان با چکیده/محل چاپ باشد.

۲ چکیده Abstract

5. Background: زمینه و شروع بحث بیان وضعیت موجود با زمان حال. A1
6. Purpose: هدف مقاله مثل A2. This paper presents... / The aim/purpose....
7. Methodology: روش/طراحی مطالعه گاهی قابل حذف. A3
8. Results: مهم‌ترین یافته‌ها با shown/indicate/found و زمان گذشته. A4
9. Conclusion: جمع‌بندی معنای نتایج/کاربرد اغلب با conclude و زمان آینده/مودال. A5

۳ مقدمه Introduction – الگوی 5M

- 1M.10: اهمیت و مرکزیت موضوع. I1
- 2M.11: پیش‌زمینه/مرور کارهای مرتبط و وضعیت فعلی دانش با ارجاع. I2
- 3M.12: شکاف/ضعف موجود اغلب با Although/However.... I3
- 4M.13: هدف مقاله و اینکه چگونه شکاف را پر می‌کند. I4
- 5M.14: نقشه/ساختار مقاله I5. The rest of the paper....

۴ جمع‌بندی و نتیجه‌گیری Conclusion

15. بازگشت به هدف/مسئله اصلی و اشاره به آن C1
16. جواب به سوال مطرح شده در انتروداکشن و پاسخ به آن معمولاً با زمان گذشته. C2

17. تفسیر/چرایی و معنی نتایج و جمع‌بندی استدلالی با خلاصه کردن مهمترین یافته ها نه بازگویی داده خام **C3**

18. محدودیت‌ها مثلاً این مقاله تنها قادر است.... **C4**

19. پیامدها/کاربردها/توصیه‌ها و مسیرهای مشخص برای کار آینده چند پیشنهاد روشن. **C5**